

CENG381 Stochastic Processes

Renewal Processes — Practice 1

Q1 (30 points). A factory installs a new lightbulb each time one burns out. Bulb lifetimes are i.i.d. with distribution X , where $E[X] = 400$ hours and $\text{Var}(X) = 160000$ hours².

- State the Elementary Renewal Theorem. What is the long-run rate of bulb replacements (bulbs per hour)?
- Approximately how many bulbs will be replaced in one year (8760 operating hours)? Using $E[X^2] = \text{Var}(X) + (E[X])^2$, compute $E[X^2]$.
- Explain the sandwich inequality used to prove the theorem:

$$N(t) / S_{\{N(t)\}} \geq N(t) / t \geq N(t) / S_{\{N(t)+1\}}$$

What does each ratio converge to by the Strong Law of Large Numbers, and how does this establish $N(t)/t \rightarrow 1/E[X]$?

Q2 (35 points). A server alternates between Busy periods (duration $\text{Exp}(\mu=2)$ hours) and Idle periods (duration $\text{Exp}(\lambda=3)$ hours). Each Busy+Idle cycle is one renewal. A reward $R = 1$ is earned per unit time while the server is Busy; $R = 0$ while Idle.

- Find $E[\text{busy period}]$ and $E[\text{idle period}]$. Hence find $E[\text{cycle length}] = E[X]$.
 - By the renewal-reward theorem, the long-run fraction of time the server is busy equals $E[\text{reward per cycle}] / E[X]$. Compute this fraction.
 - Verify your answer using the CTMC stationary distribution. Model the server as a 2-state CTMC with rate μ (Busy $\rightarrow$ Idle) and rate λ (Idle $\rightarrow$ Busy). Solve $\pi \cdot Q = 0$ to find π_{busy} .
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Q3 (35 points). A hospital bed has occupancy times following an Erlang(2, 1) distribution — equivalently, the sum of two i.i.d. $\text{Exp}(1)$ durations. $E[X] = 2$ days, $\text{Var}(X) = 2$ days², so $E[X^2] = 6$ days².

- What is the long-run bed-turnover rate (discharges per day)? In a 30-day month, how many patients do you expect to occupy this bed?
- Compute the time-average residual occupancy $Y = E[X^2] / (2 \cdot E[X])$. This is the expected remaining stay for a patient currently in the bed, sampled at a random time.

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c) Suppose instead the occupancy times were $\text{Exp}(1/2)$ (also mean 2 days). Compute $E[X^2]$ and the resulting residual occupancy. Which distribution gives a shorter residual occupancy, and why?